

09 - Pandas Library II

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1 Advanced Data Loading / 多源数据载入

【Concept / 核心概念】1

Universal Readers / 万能读取器 Pandas supports various formats to load data into DataFrames. [1]

- `pd.read_csv()`: Most common, loads comma-separated data.
- `pd.read_excel()`: Loads data from Excel sheets.
- `pd.read_html()`: Scrapes tables directly from a website URL.

2 Data Cleaning / 数据清洗（实战灵魂）

2.1 Removing Redundant Data / 删除冗余数据

【Example / 深度案例: house.csv】1

Dropping Columns When columns like 'country' or 'postcode' have the same value for all rows, they provide no information. [2, 3]

```
# Drop columns that won't be used
data = data.drop(columns=["country", "postcode"])
```

2.2 Handling NaNs / 处理缺失值

【Warning / 避坑指南】1

NaN Detection Deletion / 缺失值检测与删除

- `data.isna().sum()`: Counts how many missing values in each column. [2, 3]
- `data.dropna(inplace=True)`: Removes any row containing a missing value. Use `inplace=True` to modify the original DataFrame. [3, 4]
- `df.fillna(value)`: Replaces all null values with a specific value (e.g., the mean). [4]

2.3 Renaming Type Conversion / 重命名与类型转换

【Example / 深度案例: house.csv】1

Optimizing Column Names Types Long column names are hard to type; some data types (like objects) should be categories. [2, 3]

```
# Renaming for convenience
data.rename(columns={'rentEstimate_currentPrice': 'rentPrice'}, inplace=True)

# Type Conversion: Categorical saves memory
```

```
data['propertyType'] = data.propertyType.astype('category')
```

3 Feature Engineering / 特征工程 (进阶技巧)

3.1 Lambda Functions Assign / 匿名函数与新列创建

【Concept / 核心概念】1

lambda functions / 匿名函数 These are small, one-line functions without a name. In Pandas, they are used for mass-renaming or complex row-wise logic. [2, 5]

【Example / 深度案例: house.csv】1

Creating Flag Columns

```
# Create a boolean column 'above_mean'
data = data.assign(above_mean=lambda x: x.salePrice > x.salePrice.mean())
```

3.2 Discretization (Binning) / 数据离散化

【Concept / 核心概念】1

pd.cut() / 分箱操作 Converts a continuous numerical variable into discrete "bins" (categories). [6, 7]

【Example / 深度案例: house.csv】1

Case: Price Bins

```
# Divide salePrice into 3 categories: low, medium, high
data["salePriceBins"] = pd.cut(data["salePrice"], bins=3, labels=['low', 'medium', 'high'])
```

3.3 Encodings / 类别数据编码

- **LabelEncoder**: Converts strings into integers (e.g., 'Freehold' → 0, 'Leasehold' → 1). [2, 8]
- **One Hot Encoding (pd.get_dummies)**: Creates N new columns for N categories. Each column contains 0 or 1. [2, 8]

4 Data Exploration Aggregation / 数据探索与聚合

4.1 Pivot Table / 透视表

【Concept / 核心概念】1

Multi-dimensional Aggregation / 多维汇总 A table that groups values to aggregate items into discrete categories. [6, 7]

【Example / 深度案例: house.csv】1

Complex Analysis

```
# Analyze Sale Price Mean by number of bathrooms and price bins
# 计算不同卫浴数量在不同价格区间的平均售价
data.pivot_table(index='bathrooms', columns='salePriceBins', values='salePrice',
                  aggfunc='mean')
```

4.2 Crosstabs / 交叉表

【Example / 深度案例: house.csv】1

Frequency Table / 频率表 Quickly count how many occurrences exist for combinations of categories. [6, 7]

```
pd.crosstab(index=data["bathrooms"], columns=data["salePriceBins"])
```

5 Visualization with Pandas / 数据可视化

【Concept / 核心概念】1

Integrated Plotting / 集成绘图 Pandas provides a .plot() method that directly uses Matplotlib. [9]

【Example / 深度案例: house.csv】1

Customized Bar Plot

```
# Plotting the Pivot Table results
ax = plot_data.plot(kind='bar', rot=0, figsize=(12, 1.5), title='Bathrooms vs Price Bins')
ax.legend(loc='center', bbox_to_anchor=(0.5, -0.3), ncol=3)
```

6 Summary of Advanced Operators / 进阶操作符速查表

Operator	Description / 描述 [4, 5, 10-14]
<code>df.isna().sum()</code>	View unique missing values counts / 查看每列空值计数
<code>s.value_counts()</code>	View unique values and counts / 查看各类别出现频率
<code>df.sort_values(c, ascending=False)</code>	Sort by column in descending order / 按列降序排列
<code>pd.merge(df1, df2, how='left')</code>	SQL-style join (Left Join) / SQL 风格左连接
<code>df.groupby(c).mean()</code>	Group by column and calculate mean / 按列分组并计算均值
<code>%config InlineBackend.figure_formats</code>	Set SVG output for high resolution / 设置 SVG 高清显示